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SUBSTRATE PROCESSING APPARATUS

Abstract

A substrate processing apparatus includes a process chamber defining a process space, a first support disposed in the process space, a blocking plate supported by the first support, and a buffer extending downward from a lower surface of the blocking plate.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This patent application claims priority under 35 U.S.C. § 119 to Korean Patent Application No.10-2024-0025768, filed on Feb. 22, 2024, in the Korean Intellectual Property Office, the entire contents of which are hereby incorporated by reference.

BACKGROUND

1. Technical Field

[0002] The inventive concept relates to a substrate processing apparatus, and more specifically, relates to a substrate processing apparatus including a buffer coupled to a blocking plate.

2. Discussion of Related Art

[0003] The manufacture of a semiconductor device typically involves several processes. When processing the semiconductor device, various chemical solutions such as cleaning solutions may be used, and a drying process may be performed to remove the chemical solutions. Fluid may also be used during the drying process. An action of the fluid may cause a collapse of a semiconductor pattern of the semiconductor device, for example, due to a surface tension of the fluid. In a case that the semiconductor pattern collapses, defects may occur in the semiconductor pattern and performance of the semiconductor device may be reduced.

SUMMARY

[0004] An object of the inventive concept is to provide a substrate processing apparatus configured to reduce eddy currents inside a process chamber.

[0005] An object of the inventive concept is to provide a substrate processing apparatus configured to inhibit collapse of a semiconductor pattern.

[0006] An object of the inventive concept is to provide a substrate processing apparatus configured to increase uniformity of a flow of a supercritical fluid on a substrate.

[0007] Objects of the inventive concept are not limited to the examples mentioned herein, and other objects not mentioned will be clearly understood by those skilled in the art from the description below.

[0008] A substrate processing apparatus according to some embodiments of the inventive concept may include a process chamber defining a process space, a first support disposed in the process space, a blocking plate supported by the first support, and a buffer extending downward from a lower surface of the blocking plate.

[0009] A substrate processing apparatus according to some embodiments of the inventive concept may include a process chamber defining a process space, a first support disposed in the process space, a blocking plate supported by the first support, a second support disposed in the process space at a position spaced upward from an upper surface of the blocking plate, and a buffer coupled to a lower surface of the blocking plate, wherein the process space includes a buffer space in which the first support and the buffer are disposed, and the blocking plate defines a thickness of the buffer space.

[0010] A substrate processing apparatus according to some embodiments of the inventive concept may include a process chamber defining a process space, a blocking plate disposed in the process space, a first support supporting the blocking plate, a second support disposed in the process space and configured to support a substrate, and a buffer coupled to a lower surface of the blocking plate, wherein the process space includes a buffer space below the blocking plate, an inner surface of the process chamber includes a first lower surface defining the buffer space, and the buffer extends from the lower surface of the blocking plate toward the first lower surface.

[0011] Specific details of embodiments are included in the detailed description and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] Example embodiments will be more clearly understood from the following brief description taken in conjunction with the accompanying drawings. The accompanying drawings represent non-limiting, example embodiments as described herein.

[0013] FIG. 1 is a schematic diagram showing a substrate processing apparatus according to embodiments of the inventive concept.

[0014] FIG. 2 is a perspective view showing a substrate processing apparatus according to embodiments of the inventive concept.

[0015] FIG. 3 is a cross-sectional view showing a substrate processing apparatus according to embodiments of the inventive concept.

[0016] FIG. 4 is a perspective view showing a buffer and a blocking plate according to embodiments of the inventive concept.

[0017] FIG. 5 is a cross-sectional view showing a substrate drying device according to embodiments of the inventive concept.

[0018] FIG. 6 is a cross-sectional view showing a substrate drying device according to embodiments of the inventive concept.

DETAILED DESCRIPTION

[0019] Hereinafter, embodiments of the inventive concept will be described with reference to the attached drawings. The inventive concept may be implemented in various modifications and have various forms, and specific embodiments are illustrated in the drawings and described in detail in the text. It is to be understood, however, that the inventive concept is not intended to be limited to the particular forms disclosed, but on the contrary, is intended to cover all modifications, equivalents, and alternatives falling within the spirit and scope of the inventive concept.

[0020] The same reference numerals may refer to the same elements throughout the specification. In the drawings, the thickness, the ratio, and the dimension of the elements may be exaggerated for effective description of the technical contents.

[0021] Hereinafter, a first direction D1 and a second direction D2 may intersect each other, and a third direction D3 may cross each of the first direction D1 and the second direction D2.

[0022] FIG. 1 is a schematic diagram showing a substrate processing apparatus according to embodiments of the inventive concept.

[0023] Referring to FIG. 1, a substrate processing apparatus A may include a fluid supplier 3 and a substrate drying device 1. The substrate drying device 1 may perform a drying process. For example, a substrate W may be disposed within the substrate drying device 1 and a drying process may be performed to dry the substrate W. In this specification, the substrate W may mean a silicon (Si) substrate. However, the inventive concept is not limited thereto. The substrate drying device 1 may be implemented in various configurations.

[0024] The fluid supplier 3 may supply fluid to the substrate drying device 1. More specifically, a drying fluid supplied by the fluid supplier 3 may be injected into the substrate drying device 1. The drying fluid supplied by the fluid supplier 3 may be carbon dioxide (CO.sub.2). Carbon dioxide (CO.sub.2) injected into the substrate drying device 1 may be in a supercritical fluid (SCF) state.

[0025] The fluid supplier 3 may include a fluid source 31, a fluid line 37, a supply filter 32, a first valve 381, a condenser 33, a pump 34, a second valve 382, a tank 35, a first heater 36, and a third valve 383.

[0026] The fluid source 31 may store a drying fluid. The fluid source 31 may store the drying fluid in a gaseous state 41. The fluid source 31 may supply the drying fluid. More specifically, the fluid source 31 may store and supply the drying fluid that will become a supercritical fluid. In a case where the drying fluid is CO.sub.2, the fluid source 31 may store CO.sub.2 in a gaseous state. The drying fluid supplied from the fluid source 31 may move along the fluid line 37.

[0027] The fluid line 37 may provide a passage through which the drying fluid supplied from the

fluid source **31** flows into the substrate drying device **1**. The supply filter **32** may be disposed along the fluid line **37** or at the fluid source **31** with the fluid line **37** emanating at the supply filter **32**. The supply filter **32** may filter a foreign substance out of the drying fluid. The supply filter **32** may be omitted. The first valve **381** may control flow of the drying fluid by opening and closing a flow path between the fluid source **31** and the condenser **33**. More particularly, the first valve **381** may control flow of the drying fluid by opening and closing a flow path between the supply filter **32** and the condenser **33**.

[0028] The condenser **33** may cool the drying fluid supplied from the fluid source **31**. For example, the condenser **33** may cool gaseous CO.sub.2 supplied from the fluid source **31**. Accordingly, the drying fluid in a gaseous state may be liquefied in the condenser **33**. For example, a temperature of CO.sub.2 liquefied in the condenser **33** may be about 0 degrees Celsius (° C.) to 6° C. Additionally, a pressure of CO.sub.2 liquefied in the condenser **33** may be about 4 Megapascal (MPa) to about 6 MPa. For example, as the drying fluid is liquefied within the condenser **33**, a pressure within the fluid line **37** may increase.

[0029] The pump **34** may draw the drying fluid from the fluid source **31**. The pump **34** may draw the drying fluid from the condenser **33** disposed on a suction side of the pump **34**. The pump **34** may increase a pressure of the drying fluid within the fluid line **37**. For example, as the condenser **33** may be used to cool gaseous fluid supplied from the fluid source **31**, a pressure of the drying fluid may be increased on a discharge side of the pump **34**. For example, the pressure of CO.sub.2 liquefied in the fluid line **37** on the discharge side of the pump **34** may be about 15 MPa to about 25 MPa. Additionally, a temperature of the drying fluid passing through the pump **34** may increase. For example, the temperature of CO.sub.2 liquefied in the condenser **33** may be about 15° C. to about 25° C. as the drying fluid passes through the pump **34**. The second valve **382** may control flow of the drying fluid by opening and closing a flow path between the pump **34** and the tank **35**. For example, the second valve **382** may control flow of the drying fluid by restricting the flow path between the pump **34** and the tank **35**. The tank **35** may store the drying fluid. The tank **35** may store the drying fluid, which has been condensed into a liquid state **42** by the condenser **33** and compressed by the pump **34**.

[0030] The first heater **36** may heat the drying fluid moving along the fluid line **37**. More specifically, the first heater **36** may heat CO.sub.2 in a liquid state compressed by the pump **34**. Accordingly, liquid CO.sub.2 may become a supercritical fluid. CO.sub.2 that is heated by the first heater **36** and reaches a supercritical state may be in a high temperature and high pressure state. For example, the temperature of CO.sub.2 that has passed through the first heater **36** and reached a supercritical state may be about 60° C. to about 90° C. Additionally, the pressure of CO.sub.2 that has passed through the first heater **36** and reached a supercritical state may be at a pressure of about 15 MPa to about 25 MPa. The third valve **383** may control the flow of the drying fluid that has passed through the first heater **36** and has reached a supercritical state **43**. The third valve **383** may control flow of the drying fluid in a supercritical state by opening and closing a flow path between the first heater **36** and the substrate drying device **1**. For example, the third valve **383** may control flow of the drying fluid in a supercritical state by restricting the flow path between the first heater **36** and the substrate drying device **1**. CO.sub.2 in a supercritical state may pass through the third valve **383** and flow into the substrate drying device **1**.

[0031] FIG. 2 is a perspective view showing the substrate processing apparatus A according to embodiments of the inventive concept. FIG. 3 is a cross-sectional view showing the substrate processing apparatus A according to embodiments of the inventive concept.

[0032] Referring to FIG. 2 and FIG. 3, the substrate drying device **1** may include a process chamber **11**, a blocking plate **15**, a first support **13**, a second support **19**, a buffer **17**, a second heater **12**, a chamber driving unit **14**, a discharge tank **16**, and a fluid supplier **3**. FIG. 2 shows the substrate drying device **1** and the fluid supplier **3** for convenience.

[0033] The process chamber **11** may include a process space **11h**. For example, a drying process for

the substrate W may be performed in the process space **11h**. The process chamber **11** may have a cylindrical shape. The process chamber **11** may have a first axis **AX1** as a central axis thereof. However, a shape of the process chamber **11** is not limited thereto. The process chamber **11** may include an upper flow path **11ah** connected to an upper end portion of the process space **11h**. The process chamber **11** may provide a lower flow path **11bh** connected to a lower end portion of the process space **11h**. The upper flow path **11ah** and the lower flow path **11bh** may be connected to the process space **11h**. For example, the upper flow path **11ah** and the lower flow path **11bh** may be disposed at opposite ends of the process space **11h**. The upper flow path **11ah** may be connected to the fluid supplier **3**. The lower flow path **11bh** may be connected to the fluid supplier **3**. Fluid may flow into the process space **11h** through the upper flow path **11ah** and the lower flow path **11bh**. The lower flow path **11bh** may also be connected to the discharge tank **16**. The fluid in the process space **11h** may be discharged to the discharge tank **16** through the lower flow path **11bh**. The process chamber **11** may include a lower portion **11b** and an upper portion **11a**. A process space **11h** may be provided between the lower portion **11b** and the upper portion **11a**. The lower portion **11b** may be disposed below from the upper portion **11a**. The lower portion **11b** may move up and down (vertically). For example, the lower portion **11b** may be moved upward by operation of the chamber driving unit **14** and combined with the upper portion **11a**. However, a combination of the upper portion **11a** and the lower portion **11b** is not limited thereto. The upper portion **11a** may move up and down by operation of the chamber driving unit **14**. The upper portion **11a** may be moved up and down by the chamber driving unit **14** and combined with the lower portion **11b**. The lower portion **11b** and the upper portion **11a** may be combined to isolate the process space **11h** from the outside. The upper portion **11a** may include an upper flow path **11ah**. The lower portion **11b** may include a lower flow path **11bh**. The process space **11h** may include a buffer space **11hb**. [0034] The blocking plate **15** may be disposed in the process space **11h**. The blocking plate **15** may have a rectangular parallelepiped shape. However, a shape of the blocking plate **15** is not limited thereto.

[0035] The blocking plate **15** may be configured to support the substrate W. The blocking plate **15** may protect the lower surface of the substrate W from fluid flowing into the process chamber **11** through the lower flow path **11bh**. The blocking plate **15** may inhibit or prevent the fluid flowing into the lower flow path **11bh** from being directly sprayed onto the substrate W.

[0036] The blocking plate **15** may be supported by the first support **13**. The first support **13** extend from an outer portion of a lower surface of the blocking plate **15**. The first support **13** may be disposed in the process space **11h**. The first support **13** may be disposed in the buffer space **11hb**. A plurality of first supports **13** may be provided. For example, the plurality of first supports **13** may be arranged in a pattern around the outer portion of the lower surface of the blocking plate **15**. Hereinafter, for convenience, the plurality of first supports **13** may be treated and described as a singular component.

[0037] An area of the blocking plate **15** may be smaller than the area of the substrate W. More specifically, a diameter **DS1** of the blocking plate **15** may be about 300 millimeters (mm) or less. The diameter **DS1** of the blocking plate **15** may be larger than a diameter **DS2** of the buffer space **11hb**. A diameter of an upper surface of the buffer **17** may be less than the diameter of the blocking plate **15**.

[0038] The buffer space **11hb** may be defined by a first lower surface **111** of the process chamber **11**. An inner surface of the process chamber **11** may include the first lower surface **111** and a second lower surface **112**. A level of the first lower surface **111** may be lower than a level of the second lower surface **112**. The buffer space **11hb** may include a first space **11hb1** and a second space **11hb2**.

[0039] The first space **11hb1** may be a recessed portion of the buffer space **11hb** formed in the process space **11h**. For example, the first space **11hb1** may be a recessed space formed in the lower portion **11b** of the process chamber **11**. The first space **11hb1** may be defined by the first lower

surface **111**. For example, a height of the first space **11hb1** may be the same as a height at the recessed portion of the buffer space **11hb** formed in the process space **11h**.

[0040] The second space **11hb2** may be defined on the first space **11hb1**. The second space **11hb2** may be a space defined below the blocking plate **15** from an upper end of the first space **11hb1**.

[0041] The blocking plate **15** may be disposed above the buffer space **11hb**. A lower end portion of the first support **13** may be disposed in the first space **11hb1**. The first support **13** may extend from the first lower surface **111** to the lower surface of the blocking plate **15**. The first support **13** may be disposed in the buffer space **11hb**.

[0042] The buffer **17** may improve a uniformity of the fluid flow. The buffer **17** may be disposed in the buffer space **11hb**. The buffer **17** may extend downward from the lower surface of the blocking plate **15**. The buffer **17** may be disposed at a central portion on the lower surface of the blocking plate **15**. The buffer **17** may extend from the lower surface of the blocking plate **15** toward the first lower surface **111**. The area of the upper surface of the buffer **17** may be greater than the area of the lower surface of the buffer **17**.

[0043] A space in which fluid moves in the buffer space **11hb** may be variously changed depending on the buffer **17**. For example, the space may be reduced when a size of the buffer **17** is increased, or the space may be increased when a size of the buffer **17** is decreased.

[0044] The buffer **17** may be integrally coupled to the blocking plate **15**. However, a coupling relationship between the buffer **17** and the blocking plate **15** is not limited thereto. The buffer **17** may be a separate component from the blocking plate **15**, and may be detachable from, and attachable to the blocking plate **15**. The area of the upper surface of the buffer **17** may be smaller than the area of the lower surface of the blocking plate **15**.

[0045] A diameter **DS3** of an upper surface of the buffer **17** may be smaller than the diameter **DS2** of the buffer space **11hb**. For example, the diameter **DS3** of the upper surface of the buffer **17** may be about 30% to about 60% of the diameter **DS2** of the buffer space **11hb**. However, the diameter **DS3** of the upper surface of the buffer **17** is not limited thereto. In a case that the diameter **DS3** of the upper surface of the buffer **17** is smaller than the diameter **DS1** of the blocking plate **15** and the diameter **DS2** of the buffer space **11hb**, a rate of the fluid flow from the lower flow path **11bh** may be decreased.

[0046] In a case that the area of the upper surface of the buffer **17** is greater than the area of the lower surface of the buffer **17**, a height of the buffer space **11hb** increases away from the first axis **AX1** of the process chamber **11**. More specifically, when the fluid flowing in from the lower flow path **11bh** collides with the buffer **17** and flows toward the first support **13**, the buffer space **11hb** through which the fluid moves may have an increasing height and the fluid movement rate may decrease. For example, the buffer space **11hb** through which the fluid moves may expand away from the first axis **AX1** and the fluid movement rate may decrease.

[0047] A thickness of the buffer **17** may be smaller than a thickness of the buffer space **11hb**. For example, a thickness of the buffer **17** may be about 60% or less of a thickness of the buffer space **11hb**. The thickness of the buffer space **11hb** may refer to a distance from the first lower surface **111** to the lower surface of the blocking plate **15**. For example, the blocking plate **15** may define a thickness of the buffer space **11hb**. However, the thickness of the buffer **17** is not limited thereto.

[0048] The second support **19** may be disposed in the process space **11h**. The second support **19** may support the substrate **W**. The second support **19** may place the substrate **W** at a position spaced upward from the blocking plate **15**. The second support **19** may extend upward from the upper surface of the blocking plate **15**. In other words, a level of an upper end portion of the second support **19** may be higher than a level of an upper surface of the blocking plate **15**. A level of a lower surface of the second support **19** may be higher than a level of a lower surface of the blocking plate **15**. A plurality of second supports **19** may be provided. For example, the plurality of second supports **19** may be arranged in a pattern on the upper surface of the blocking plate **15**. Hereinafter, for convenience, the plurality of second supports **19** will be treated and described as a

singular component. A height of the second support **19** in the first direction **D1** may be less than a height of the first support **13** in the first direction **D1**.

[0049] The second heater **12** may be disposed in the process chamber **11**. The second heater **12** may be thermally coupled to the process chamber **11**. The second heater **12** may heat the process space **11h**. By heating the process space **11h**, the fluid flowing into the process space **11h** may be maintained in a supercritical state.

[0050] FIG. **4** is a perspective view showing the buffer **17** and the blocking plate **15** according to embodiments of the inventive concept. FIG. **5** is a cross-sectional view showing the substrate drying device **1** according to embodiments of the inventive concept. FIG. **6** is a cross-sectional view showing the substrate drying device **1** according to embodiments of the invention.

[0051] Referring to FIG. **4**, the area of a lower surface of the buffer **17** may be smaller than the area of an upper surface of the buffer **17**. The lower surface of the buffer **17** and the upper surface of the buffer **17** may be connected to a side surface of the buffer **17**. The lower surface of the buffer **17** and the upper surface of the buffer **17** may be connected to each other by the side surface of the buffer **17**. An angle formed between the side surface of the buffer **17** and the lower surface of the blocking plate **15** may be about 60 degrees or less. A shape of the buffer **17** is not limited thereto. For example, referring to FIG. **5**, the buffer **17** may have a rectangular parallelepiped shape.

Referring to FIG. **6**, an outer surface of the buffer **17** may include a curved surface. The buffer **17** may have a hemispherical shape. However, the shape of the buffer **17** is not limited to the shape of FIG. **4**, FIG. **5**, and FIG. **6**. For example, the buffer **17** may have a concave shape. The buffer **17** may have a different shape that may reduce flow rate of the fluid when the fluid flows into the buffer space **11hb**.

[0052] According to the substrate processing apparatus according to embodiments of the inventive concept, flow uniformity of the fluid flowing on an upper surface of the substrate may be increased. In the case that the buffer is coupled to the lower surface of the blocking plate, a volume of space through which fluid moves may increase away from a center portion of the process chamber **11**. The height of the space at the center portion of the process chamber **11** may be different from the height of the space in which the fluid moves outside the buffer space. For example, the volume of the space may have a cross section that increase as the fluid moves from the center portion of the process chamber to outside the buffer space. As the fluid moves from inside the buffer space to outside the buffer space, a volume of the space in which the fluid moves may increase. As the volume of space in which the fluid moves increases, the flow rate of the fluid may decrease as the fluid moves from the center portion of the buffer space to outside the buffer space. The first support may inhibit the flow rate of the fluid, and the flow rate of the fluid may decrease. For example, as the fluid collides with the first support, the fluid flow rate may be slowed and eddy currents in the fluid may be decreased. By reducing eddy currents in the fluid occur, the flow of fluid on the substrate may become more uniform. Further, by reducing eddy currents in the fluid, stress may be on the semiconductor pattern may be reduced. As the flow rate of the fluid is reduced by the buffer, a collapse of the semiconductor pattern and deterioration of quality due to the fluid may be inhibited or prevented.

[0053] According to the substrate processing apparatus of the inventive concept, an eddy current inside the process chamber may be reduced or prevented.

[0054] According to the substrate processing apparatus of the inventive concept, the collapse of the semiconductor pattern may be inhibited or prevented.

[0055] According to the substrate processing apparatus of the inventive concept, flow uniformity of the supercritical fluid on the substrate may be increased.

[0056] Effects of the inventive concept are not limited to the effects mentioned herein, and other effects not mentioned will be clearly understood by those skilled in the art from the description below.

[0057] While embodiments are described above, a person skilled in the art may understand that

many modifications and variations are made without departing from the spirit and scope of the inventive concept defined in the following claims. Accordingly, example embodiments of the inventive concept should be considered in all respects as illustrative and not restrictive, with the spirit and scope of the inventive concept being indicated by the appended claims.

Claims

1. A substrate processing apparatus comprising: a process chamber defining a process space; a first support disposed in the process space; a blocking plate supported by the first support; and a buffer extending downward from a lower surface of the blocking plate.
2. The substrate processing apparatus of claim 1, wherein the process space includes a buffer space defined below the blocking plate, wherein the buffer space includes: a first space in which a lower end portion of the first support is disposed on the process chamber; and a second space on the first space, wherein the buffer is disposed in the buffer space.
3. The substrate processing apparatus of claim 2, wherein a diameter of an upper surface of the buffer is about 30% to about 60% of a diameter of the buffer space.
4. The substrate processing apparatus of claim 2, wherein a thickness of the buffer is about 60% or less of a thickness of the buffer space.
5. The substrate processing apparatus of claim 1, wherein the process chamber includes: an upper flow path connected to an upper end portion of the process space; and a lower flow path connected to a lower end portion of the process space.
6. The substrate processing apparatus of claim 1, wherein the buffer is disposed at a center portion of the blocking plate, and wherein the first support extends downward from the lower surface of the blocking plate and disposed at an outer portion of the blocking plate.
7. The substrate processing apparatus of claim 1, further comprising a second support disposed in the process space and configured to support a substrate, wherein the second support is disposed a position spaced upward from an upper surface of the blocking plate.
8. The substrate processing apparatus of claim 1, further comprising a second support disposed in the process space and configured to support a substrate, wherein the second support extends upward from an upper surface of the blocking plate.
9. The substrate processing apparatus of claim 1, wherein an area of a lower surface of the buffer is smaller than an area of an upper surface of the buffer, wherein the lower surface of the buffer and the upper surface of the buffer are connected to a side surface of the buffer, and wherein an angle formed between the side surface of the buffer and the lower surface of the blocking plate is about 60 degrees or less.
10. The substrate processing apparatus of claim 1, wherein the buffer has a rectangular parallelepiped shape.
11. The substrate processing apparatus of claim 1, wherein an outer surface of the buffer includes a curved surface.
12. The substrate processing apparatus of claim 1, wherein the blocking plate has a diameter of about 300 millimeters or less, and a diameter of an upper surface of the buffer is less than the diameter of the blocking plate.
13. A substrate processing apparatus comprising: a process chamber defining a process space; a first support disposed in the process space; a blocking plate supported by the first support; a second support disposed in the process space at a position spaced upward from an upper surface of the blocking plate; and a buffer coupled to a lower surface of the blocking plate, wherein the process space includes a buffer space in which the first support and the buffer are disposed, and wherein the blocking plate defines a thickness of the buffer space.
14. The substrate processing apparatus of claim 13, wherein a diameter of the blocking plate is greater than a diameter of the buffer space, and wherein a diameter of an upper surface of the buffer

is about 30% to about 60% of a diameter of the buffer space.

15. The substrate processing apparatus of claim 13, wherein a thickness of the buffer is about 60% or less of a thickness of the buffer space.

16. The substrate processing apparatus of claim 13, wherein the second support extends upward from the upper surface of the blocking plate and is configured to support a substrate.

17. A substrate processing apparatus comprising: a process chamber defining a process space; a blocking plate disposed in the process space; a first support supporting the blocking plate; a second support disposed in the process space and configured to support a substrate; and a buffer coupled to a lower surface of the blocking plate, wherein the process space includes a buffer space below the blocking plate, wherein an inner surface of the process chamber includes a first lower surface defining the buffer space, and wherein the buffer extends from the lower surface of the blocking plate toward the first lower surface.

18. The substrate processing apparatus of claim 17, wherein an area of an upper surface of the buffer is smaller than an area of the lower surface of the blocking plate.

19. The substrate processing apparatus of claim 17, wherein the first support and the buffer are disposed in the buffer space.

20. The substrate processing apparatus of claim 17, wherein a diameter of an upper surface of the buffer is about 30% to about 60% of a diameter of the buffer space, and wherein a thickness of the buffer is about 60% or less of a thickness of the buffer space.
